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Paint sprayer

Abstract

A paint sprayer includes a cart assembly supporting a motor and a fluid intake conduit. The cart assembly has a frame with a pair of spaced-apart legs for supporting the cart assembly in an upright position, a telescoping handle movable between a retracted handle position and an extended handle position, an upper bracket projecting from the telescoping handle and a telescoping support movable between a retracted support position and an extended support position. The telescoping support supports the cart assembly in an inclined position between the upright position and a horizontal position of the cart assembly. The cart assembly has a pair of lower wheels for rolling the cart assembly when tilted back from the upright position and at least one upper wheel disposed on the telescoping handle for rolling the cart assembly along with the pair of lower wheels when in the horizontal position.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation-in Part application of and claims priority under 35 U.S.C. § 120 and the benefit of co-pending U.S. Design patent application No. 29/882,620 filed on Jan. 17, 2023, the disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION
Field of the Invention
(1) The invention relates to a paint sprayer, in particular a portable paint sprayer and a method for operating and transporting a portable paint sprayer.
The Prior Art
(2) Various paint sprayer devices are known. It is known to mount such paint sprayers to a wheeled frame for moving the paint sprayer to and from a location where it is to be operated. Due to their large, bulky size and heavy weight (paint sprayers may weigh between 50 and 200 pounds, even without an associated hose and spray gun) existing paint sprayers typically require two persons to load and/or unload the paint sprayer into/out of a vehicle, for example a truck bed, van floor or the like.
(3) Moreover, operators often need to employ loading ramps to roll the bulky, heavy paint sprayer to and from the surface of the vehicle which is elevated relative to the floor or ground. Such loading ramps take up additional space in the vehicle when being transported with the paint sprayer and require additional room behind the vehicle when being used. Because storage space in the vehicle and parking area size may be limited, it would be advantageous to eliminate the need for ramps when loading and unloading the paint sprayer.
(4) Accordingly, a need exists for a paint sprayer arrangement which can be easily loaded and unloaded into and out of a vehicle by one person without the need for ramps or other equipment.
SUMMARY OF THE INVENTION
(5) The invention relates to a paint sprayer, in particular a portable paint sprayer and a method for operating and transporting a portable paint sprayer.
(6) A paint sprayer according to an aspect of the invention includes a motor having a gear box and a

fluid intake conduit coupled to the motor. The fluid intake conduit includes an inlet end configured to be submerged in a container of paint. The paint sprayer further includes an outlet fitting in fluid communication with the fluid intake conduit and a cart assembly supporting the motor and the fluid intake conduit. The cart includes a frame having a pair of spaced-apart legs disposed at a lower portion of the frame and configured to support the cart assembly in an upright position of the cart assembly; a telescoping handle disposed at an upper portion of the frame and movable between a retracted handle position and an extended handle position; an upper bracket projecting from the telescoping handle and configured to receive a first portion of a coiled flexible hose; and a telescoping support disposed between the lower portion of the frame and the upper portion of the frame and movable between a retracted support position and an extended support position. The telescoping support is configured to support the cart assembly in an inclined position of the cart assembly between the upright position of the cart assembly and a horizontal position of the cart assembly. The telescoping support is also configured to receive a second portion of the coiled flexible hose.
(7) The cart assembly further includes a pair of lower wheels disposed at the lower portion of the frame and configured for rolling the cart assembly when tilted back from the upright position and at least one upper wheel disposed on the telescoping handle and configured for rolling the cart assembly along with the pair of lower wheels when in the horizontal position.
(8) According to a further aspect of the invention, the motor is an electric motor.
(9) According to a further aspect of the invention, the at least one upper wheel is a pair of upper wheels.
(10) According to a further aspect of the invention, the paint sprayer has a respective grip handle disposed on each of the pair of spaced-apart legs.
(11) According to a further aspect of the invention, the telescoping handle has a locking mechanism configured for locking the telescoping handle in the retracted handle position and the extended handle position.
(12) According to a further aspect of the invention, the locking mechanism for the telescoping handle includes a plurality of spaced apart openings and a spring biased button.
(13) According to a further aspect of the invention, the telescoping support has a locking mechanism configured for locking the telescoping support in the retracted support position and the extended support position.
(14) According to a further aspect of the invention, the locking mechanism for the telescoping support includes a plurality of spaced apart openings and a spring biased button.
(15) A method for operating and transporting a paint sprayer according to an aspect of the invention includes the step of providing a paint sprayer including a motor having a gear box, a fluid intake conduit coupled to the motor, an outlet fitting in fluid communication with the fluid intake conduit, a cart assembly supporting the motor and the fluid intake conduit, the cart assembly including a frame having a pair of spaced-apart legs disposed at a lower portion of the frame, a telescoping handle disposed at an upper portion of the frame, an upper bracket projecting from the telescoping handle and a telescoping support disposed between the lower portion of the frame and the upper portion of the frame, the cart assembly further including a pair of lower wheels disposed at the lower portion of the frame and at least one upper wheel disposed on the telescoping handle.
(16) The method further includes the step of moving the paint sprayer to a selected location by grasping the telescoping handle, tilting the cart assembly back from an upright position of the cart assembly and rolling the cart assembly on the pair of lower wheels.
(17) The method further includes the step extending the telescoping support into an extended support position and tilting the cart assembly back such that the paint sprayer is supported by the telescoping support in an inclined position of the cart assembly between the upright position of the cart assembly and a horizontal position of the cart assembly for placing a container of paint under an inlet end of the fluid intake conduit.

(18) The method further includes the step of retracting the telescoping support into a retracted support position and loading the paint sprayer onto a support surface of a vehicle by positioning the at least one upper wheel on the support surface of the vehicle, lifting the cart assembly into the horizontal position and rolling the cart assembly onto the support surface on the pair of lower wheels and the at least one upper wheel.
(19) In a method according to a further aspect of the invention, the step of lifting the cart assembly into the horizontal position includes grasping each of a respective grip handle disposed on each of the pair of spaced-apart legs.
(20) In a method according to a further aspect of the invention, the step of extending the telescoping support into the extended support position includes locking the telescoping support in the extended support position with a locking mechanism.
(21) In a method according to a further aspect of the invention, the step of retracting the telescoping support into the retracted support position includes locking the telescoping support in the retracted support position with a locking mechanism.
(22) A method according to a further aspect of the invention includes the steps of supporting a coiled flexible hose between the upper bracket and the telescoping support with the telescoping handle in an extended handle position; moving the telescoping handle from the extended handle position to a retracted handle position; and removing the coiled flexible hose from the upper bracket and the telescoping support.
(23) In a method according to a further aspect of the invention, the inclined position of the cart assembly is in the range of 45 to 60 degrees from the upright position of the cart assembly.
(24) In a method according to a further aspect of the invention, the support surface of the vehicle is a truck bed or a floor of a van.
(25) In a method according to a further aspect of the invention, the step of loading the paint sprayer onto the support surface of the vehicle is completed without the use of a ramp.
(26) An advantage of a paint sprayer and a method for operating and transporting a paint sprayer according to embodiments of the invention is the provision of a paint sprayer arrangement which can be easily loaded and unloaded into and out of a vehicle by one person without the need for ramps or other equipment.
(27) Moreover, a paint sprayer and a method for operating and transporting a paint sprayer according to embodiments of the invention can be easily loaded and unloaded into and out of a vehicle as a complete paint spraying rig, including the paint sprayer and its associated hose(s) and paint spray gun(s)
(28) Another advantage of a paint sprayer and a method for operating and transporting a paint sprayer according to embodiments of the invention is that a coiled flexible hose may be securely supported between an upper support bracket coupled to a locking telescoping handle and a lower locking telescoping support with the telescoping handle in an extended handle position and the coiled flexible hose can be easily removed from the two supports when the telescoping handle is moved from the extended handle position to a retracted handle position.
(29) Another advantage of a paint sprayer and a method for operating and transporting a paint sprayer according to embodiments of the invention is that a container of paint, such as a five gallon bucket of paint, can be easily placed in close proximity to the inlet end of a fluid intake conduit of the paint sprayer by extending a telescoping support into an extended support position and tilting the cart assembly back such that the paint sprayer is supported by the telescoping support in an inclined position of the cart assembly between an upright position and a horizontal position, for example at an angle in the range of 45 to 60 degrees from the upright position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Other advantages, benefits and features of the present invention will become apparent from the following detailed description considered in connection with the accompanying drawings. It is to be understood, however, that the drawings are designed as an illustration only and not as a definition of the limits of the invention.

(2) In the drawings, wherein similar reference characters denote similar elements throughout the several views:

(3) FIG. **1** shows a perspective view of a paint sprayer according to an embodiment of the invention;

(4) FIG. **2** shows a side view of the paint sprayer shown in FIG. **1**;

(5) FIG. **3** shows a perspective view of a paint sprayer according to another embodiment of the invention;

(6) FIG. **4** shows a side view of the paint sprayer shown in FIG. **3**;

(7) FIG. **5** shows the paint sprayer tilted back from an upright position for moving the paint sprayer to a selected location;

(8) FIG. **6** shows the paint sprayer tilted back and supported by a telescoping support for placing a container of paint under an inlet end of the fluid intake conduit;

(9) FIG. **7** shows the paint sprayer being loaded onto a support surface of a vehicle;

(10) FIG. **8** shows the paint sprayer supporting a coiled flexible hose between the upper bracket and the telescoping support with the telescoping handle in an extended handle position; and

(11) FIG. **9** shows the telescoping handle in a retracted handle position for removing the coiled flexible hose from the upper bracket and the telescoping support.

DETAILED DESCRIPTION OF THE DRAWINGS

(12) Referring now in detail to the drawings, and in particular FIG. **1**, a shows a perspective view of a paint sprayer **1** according to an embodiment of the invention and FIG. **2** shows a side view of the paint sprayer **1** shown in FIG. **1**. The paint sprayer **1** includes a motor **2** having a gear box. The motor **2** may be an electric motor or any suitable motor for generating pressure to spray the paint.

(13) The motor **2** drives gears that are coupled to a connecting which moves a piston up and down inside a fluid section of the paint sprayer. The speed and/or/cycles (input) of this movement creates a desired output which translates into the paint spray pressure delivered for a desired spray pattern performance.

(14) The motor **2** may be coupled to a control box **21** having a pressure control knob **22** for controlling the motor's speed. A precision motor control mechanism (PMC) may match the motor speed to a desired pressure and selected spray tip size and eliminate pulsation for all pressures and tip sizes.

(15) Paint sprayer **1** may also include a filter system **3** including a filter housing **31** and a paint screening filter **32** that prevents dirt or other particles from moving to the spray tip where they could clog the tip and impede spraying performance.

(16) One or more outlet fittings **33** (hose fittings) may be provided at an outlet portion of the filter system **3**. The outlet fitting or fittings **33** are configured to connect a flexible high pressure hose **200** (See FIGS. **8** and **9**) so that the filtered paint flows through the flexible hose **200** to a spray gun (not shown). The flexible hoses **200** allow painters to spray at longer distances from the paint sprayer **1** so that the paint sprayer does not have to be relocated. The flexible hoses **200** typically have lengths of fifty to three hundred feet. The provision of multiple outlet fittings **33** allow two or more operators to spray simultaneously, each using an individual hose **200** coupled to a single paint sprayer unit **1**.

(17) A pressure relief valve **34** may be provided on the filter system **3**. Pressure relief valve **34** may be a mechanical valve that allows for the paint to be primed through the paint sprayer **1**. After priming, the painter switches the pressure relief valve **34** to a spray position when ready to spray

under high pressure. The pressure relief valve **34** can also be used to relieve the high pressure and allow for cleaning paint from the unit.

(18) A fluid intake conduit **4** is coupled to the motor **2**. The fluid intake conduit **4** is in fluid communication with the outlet fitting or fittings **33** and has an inlet end **41** configured to be submerged in a container of paint **100** (See FIG. **6**), for example a conventional five gallon bucket of paint. The fluid intake conduit **4** or a portion thereof is submerged into container of paint **100** for direct intake of the paint. The fluid intake conduit **4** functions to siphon the paint which is pressurized to be atomized and sprayed onto a surface by means of a paint spray gun with spray tip (not shown).

(19) Paint sprayer **1** may include a pail hook **5** disposed on a front portion of the unit. The pail hook **5** is used to attach the handle **101** of a paint container **100**, such as the handle of a five gallon paint bucket. When moving the paint sprayer to another location, the paint container **100** can be attached to the unit by hanging the handle **101** on the pail hook **5**, tilting the paint sprayer **1** back on its lower wheels **6**, thereby lifting the paint container **100** of the floor or ground. In this way, the paint container **100** can move with the rolling paint sprayer **1** to another location and the painter does not need to remove the fluid intake conduit **4** out of the paint container **100** when moving the paint sprayer **1** around a jobsite.

(20) Paint sprayer **1** may also have a power cord hook **7** which is used to support a coiled electrical power cord when storing and transporting the paint sprayer **1**.

(21) Paint sprayer **1** further includes a cart assembly **8** which supports the motor **2** and the fluid intake conduit **4**. The cart assembly **8** includes a frame **9**.

(22) Frame **9** includes a pair of spaced-apart legs disposed **91** at a lower portion of the frame **9** and configured to support the cart assembly **8** in an upright position of the cart assembly, as shown for example in FIGS. **1-4**, **8** and **9**. Each of the spaced-apart legs **91** may have a respective grip handle **95** disposed thereon. Grip handles **95** are configured to facilitate lifting the paint sprayer unit **1** and tilting it backward from an upright position to a horizontal position for easy loading and unloading into a vehicle **300**, as shown in FIG. **7**.

(23) Frame **9** also has a telescoping handle **92** disposed at an upper portion of the frame **9** and movable between a retracted handle position, shown for example in broken line in FIGS. **1-4** and an extended handle position, shown for example in solid line in FIGS. **1-4**. FIG. **8** shows telescoping handle **92** in the extended handle position for supporting a coiled flexible hose **200** and FIG. **9** shows telescoping handle **92** in the retracted handle position for removal of the coiled flexible hose **200**.

(24) Telescoping handle **92** may include a locking mechanism configured for locking the telescoping handle **92** in at least the retracted handle position and the extended handle position. The locking mechanism can include a plurality of spaced apart openings **96** and a spring biased button **97**. Spring biased button **97** is configured to engage a selected one of the spaced apart openings in order to lock the telescoping handle **92** in a desired position, for example in an extended position for rolling the paint sprayer unit **1** to a location for use, as shown in FIG. **5** or a retracted position for transporting or storing the paint sprayer unit **1** or for removing the coiled flexible hose **200** as shown in FIG. **9**.

(25) Frame **9** has an upper bracket **93** projecting from the telescoping handle **92** and configured to receive a first or upper portion **201** of a coiled flexible hose **200**.

(26) Frame **9** also has a telescoping support **94** disposed between the lower portion of the frame **9** and the upper portion of the frame **9** and movable between a retracted support position, shown for example in solid line in FIGS. **1-4** and an extended support position, shown for example in broken line in FIGS. **1-4**. Telescoping support **94** is configured to support the cart assembly **8** in an inclined position of the cart assembly **8** between the upright position of the cart assembly and a horizontal position of the cart assembly **8** as shown in FIG. **6**. Telescoping support **94** is moved to its retracted position when loading and unloading the paint sprayer **1**, which allows the unit to roll

on its at least one upper wheel **81** and lower wheels **6**.

(27) Telescoping support **94** may include a locking mechanism configured for locking the telescoping support **94** in at least in the retracted support position and the extended support position. The locking mechanism can be substantially the same as described above for telescoping handle **92** and can include a plurality of spaced apart openings **98** and a spring biased button **99** configured to engage a selected one of the spaced apart openings **98** in order to lock the telescoping support **94** in a desired position. As shown in FIG. **6**, telescoping support **94** may be locked in a position to support paint sprayer **1** so that a container **100** of paint can be easily placed under an inlet end **41** of the fluid intake conduit **5**.

(28) Telescoping support **94** is configured to receive a second or lower portion **202** of the coiled flexible hose **200** as shown in FIGS. **8** and **9**.

(29) Cart assembly **8** includes a pair of lower wheels **6** disposed at the lower portion of the frame **9** and configured for rolling the cart assembly **8** when tilted back from the upright position as shown in FIG. **5**. At least one upper wheel **81** is disposed on the telescoping handle **92** and is configured for rolling the cart assembly **8** along with the pair of lower wheels **6** when the paint sprayer **1** is in a horizontal position. The at least one upper wheel **81** may be a single upper wheel **81** which is centered on the cross member of the telescoping handle **92** as shown in FIGS. **1** and **2**, a pair of upper wheels **81** as shown in FIGS. **3** and **4**, or any suitable number of upper wheels. Preferably the one or more upper wheels **81** are smaller in diameter than lower wheels **6**.

(30) In order to operate and transport a paint sprayer **1** according to an aspect of the invention, a paint sprayer **1** having a motor **2** with a gear box **21**, a fluid intake conduit **4** coupled to the motor **2**, an outlet fitting **33** in fluid communication with the fluid intake conduit **4** and a cart assembly **8** supporting the motor **2** and the fluid intake conduit **4** is provided. The cart assembly **8** includes a frame **9** having a pair of spaced-apart legs **91** disposed at a lower portion of the frame **9**, a telescoping handle **92** disposed at an upper portion of the frame **9**, an upper bracket **93** projecting from the telescoping handle **92** and a telescoping support **94** disposed between the lower portion of the frame **9** and the upper portion of the frame **9**. The cart assembly **8** further includes a pair of lower wheels **6** disposed at the lower portion of the frame **9** and at least one upper wheel **81** disposed on the telescoping handle **92**.

(31) As illustrated in FIG. **5**, the paint sprayer **1** can be moved to a selected location by grasping the telescoping handle **92**, tilting the cart assembly **8** back from an upright position of the cart assembly **8** and rolling the cart assembly **8** on the pair of lower wheels **6**.

(32) As illustrated in FIG. **6**, the telescoping support **94** may be extended and locked into an extended support position and the cart assembly **8** tilted back such that the paint sprayer **1** is supported by the telescoping support **94** in an inclined position of the cart assembly **8** between the upright position of the cart assembly **8** and a horizontal position of the cart assembly **8** for placing a container of paint **100** under an inlet end **41** of the fluid intake conduit **4**. The telescoping support **94** may be configured to support the cart assembly **8** in an inclined position is in the range of 45 to 60 degrees from the upright position of the cart assembly (See FIG. **6**, angle β).

(33) The telescoping support **94** may also be retracted and locked into a retracted support position the paint sprayer **1** loaded the paint sprayer onto a support surface **301** of a vehicle **300** by positioning the at least one upper wheel **81** on the support surface **301** of the vehicle **300**, lifting the cart assembly **8** into the horizontal position and rolling the cart assembly **8** onto the support surface **301** on the pair of lower wheels **6** and the at least one upper wheel **81**. The support surface **301** of the vehicle **300** can be a truck bed, a floor of a van, a trailer bed or any other suitable surface. In this way, loading and unloading of the paint sprayer **1** can be accomplished by a single operator without the use of a ramps or other equipment. The loading and unloading operation may be facilitated by grasping respective grip handles **95** provided on each of the spaced apart legs **91** of the frame, as shown in FIG. **7**.

(34) The telescoping support **94** is preferably locked in a selected extended support position or

retracted position using a locking mechanism, for example a plurality of spaced apart openings **98** which are engaged by a spring biased button **99**.
(35) A coiled flexible hose **200** adapted to be coupled to an outlet fitting **33** at one send and a paint spray gun at the other end can be supported between the upper bracket **93** and the telescoping support **94** with the telescoping handle in an extended handle position as shown in FIG. **8**. The telescoping handle **92** is then moved from the extended handle position to a retracted handle position as shown in FIG. **9**. This allows the coiled flexible hose **200** to be easily removed from the upper bracket **93** and the telescoping support **94** as shown in FIG. **9**.
(36) Although a number of embodiments of the present invention have been shown and described, it is obvious that many changes and modifications may be made thereunto without departing from the spirit and scope of the invention.

Claims

1. A paint sprayer comprising: a motor having a gear box; a fluid intake conduit coupled to the motor, the fluid intake conduit comprising an inlet end configured to be submerged in a container of paint; an outlet fitting in fluid communication with the fluid intake conduit; and a cart assembly supporting the motor and the fluid intake conduit; wherein the cart assembly comprises a frame; the frame comprising: a pair of spaced-apart legs disposed at a lower portion of the frame and configured to support the cart assembly in an upright position of the cart assembly; a telescoping handle disposed at an upper portion of the frame and movable between a retracted handle position and an extended handle position; an upper bracket projecting from the telescoping handle and configured to receive a first portion of a coiled flexible hose; and a telescoping support disposed between the lower portion of the frame and the upper portion of the frame and movable between a retracted support position and an extended support position, wherein the telescoping support is configured to support the cart assembly in an inclined position of the cart assembly between the upright position of the cart assembly and a horizontal position of the cart assembly and wherein the telescoping support is configured to receive a second portion of the coiled flexible hose; and wherein the cart assembly further comprises a pair of lower wheels disposed at the lower portion of the frame and configured for rolling the cart assembly when tilted back from the upright position and at least one upper wheel disposed on the telescoping handle and configured for rolling the cart assembly along with the pair of lower wheels when in the horizontal position.
2. The paint sprayer according to claim 1, wherein the motor comprises an electric motor.
3. The paint sprayer according to claim 1, wherein the at least one upper wheel comprises a pair of upper wheels.
4. The paint sprayer according to claim 1, further comprising a respective grip handle disposed on each of the pair of spaced-apart legs.
5. The paint sprayer according to claim 1, wherein the telescoping handle comprises a locking mechanism configured for locking the telescoping handle in the retracted handle position and the extended handle position.
6. The paint sprayer according to claim 5, wherein the locking mechanism comprises a plurality of spaced apart openings and a spring biased button.
7. The paint sprayer according to claim 1, wherein the telescoping support comprises a locking mechanism configured for locking the telescoping support in the retracted support position and the extended support position.
8. The paint sprayer according to claim 7, wherein the locking mechanism comprises a plurality of spaced apart openings and a spring biased button.
9. A method for operating and transporting a paint sprayer, the method comprising the steps of: providing a paint sprayer comprising a motor having a gear box, a fluid intake conduit coupled to the motor, an outlet fitting in fluid communication with the fluid intake conduit, a cart assembly

supporting the motor and the fluid intake conduit, the cart assembly comprising a frame having a pair of spaced-apart legs disposed at a lower portion of the frame, a telescoping handle disposed at an upper portion of the frame, an upper bracket projecting from the telescoping handle and a telescoping support disposed between the lower portion of the frame and the upper portion of the frame, the cart assembly further comprising a pair of lower wheels disposed at the lower portion of the frame and at least one upper wheel disposed on the telescoping handle; moving the paint sprayer to a selected location by grasping the telescoping handle, tilting the cart assembly back from an upright position of the cart assembly and rolling the cart assembly on the pair of lower wheels; extending the telescoping support into an extended support position and tilting the cart assembly back such that the paint sprayer is supported by the telescoping support in an inclined position of the cart assembly between the upright position of the cart assembly and a horizontal position of the cart assembly for placing a container of paint under an inlet end of the fluid intake conduit; and retracting the telescoping support into a retracted support position and loading the paint sprayer onto a support surface of a vehicle by positioning the at least one upper wheel on the support surface of the vehicle, lifting the cart assembly into the horizontal position and rolling the cart assembly onto the support surface on the pair of lower wheels and the at least one upper wheel.
10. The method according to claim 9, wherein the step of lifting the cart assembly into the horizontal position comprises grasping each of a respective grip handle disposed on each of the pair of spaced-apart legs.
11. The method according to claim 9, wherein the step of extending the telescoping support into the extended support position comprises locking the telescoping support in the extended support position with a locking mechanism.
12. The method according to claim 9, wherein the step of retracting the telescoping support into the retracted support position comprises locking the telescoping support in the retracted support position with a locking mechanism.
13. The method according to claim 9, further comprising the steps of: supporting a coiled flexible hose between the upper bracket and the telescoping support with the telescoping handle in an extended handle position; moving the telescoping handle from the extended handle position to a retracted handle position; and removing the coiled flexible hose from the upper bracket and the telescoping support.
14. The method according to claim 9, wherein the inclined position of the cart assembly is in the range of 45 to 60 degrees from the upright position of the cart assembly.
15. The method according to claim 9, wherein the support surface of the vehicle comprises a truck bed or a floor of a van.
16. The method according to claim 9, wherein the step of loading the paint sprayer onto the support surface of the vehicle is completed without the use of a ramp.
